

ANSH MITTAL

Mountain View, CA | (213) 573-9188 | manshm1811@gmail.com | linkedin.com/in/mittalansh/ | github.com/AnshMittal1811 |
medium.com/@anshm1811996 | https://anshmittal1811.github.io/

SUMMARY

5+ yrs of ML experience bridging 3D Computer Vision and LLM optimization, spanning foundational research to enterprise production. Proven track record of shipping spatial AI pipelines and architecting large-scale agentic AI systems. Skilled in translating cutting-edge prototypes (NeRFs, Gaussian Splats) and fine-tuned Language models into scalable, production-ready workflows.

EXPERIENCE

Guidewire Software Inc | *San Mateo, CA*

April 2024–Present

• Machine Learning Engineer II - GenAI, Research

- Automated prompt-prefix tuning for OCR–RAG–LLMs using RLAIF, Bayesian Optimization (for LLM hyperparam-tuning), and CMAES (for RAG hyperparam-tuning) across 5 lines-of-business (avg score 0.875 (Levenshtein, F1-score, ROUGE-L))
- Finetuned and Deployed Qwen-3-4B claim summarization QLoRA adapter with 0.76 score (6% ↑) (ROUGE-L, BERTScore, Domain-based) based on comparative analysis with GEMMA 3-4B QLoRA adapter and base models
- Innovated auto-annotation pipeline for community outline segmentation with hazard-based SatelliteMAE Optical/SAR modelling (Avg Dice Coef.: 0.92 (10% ↑)) associated with late fusion for public safety buildings (from news text/images)
- Developed Agentic AI workflow with tool calling, Chain-of-Thoughts, Tree-of-Thoughts, Graph-of-Thoughts (branched SFT DocLLM, OCR-RAG-LLM, BDA) to reduce False Positive Rates in Year Loss Tables for Cyber Risk Modeling
- Architected Digital Twin simulation infrastructure to model complex cyber risks on network infrastructure, analyzing phase shift signals and packet captures to simulate and predict multi-stage attack vectors
- Built batch/on-demand feature store and pipelines with S3, Step Functions, DAGs, ECS services using Teamcity CI/CD with CodeLLAMA (HCL generation (ev_{acc} :81% (11% ↑))) & integrated Mann-Whitney U gate/z-score temporal drift monitoring

Remix Inc | *Palo Alto, CA*

September 2023–December 2023

• Senior Computer Vision Engineer (Founding)

- Integrated 3D dynamic view reconstruction with LeRF-joint training and human Gaussian-splatting for semantically 3D-aware panoramic systems, achieving 38ms latency in real-time novel scenes
- Compiled custom Gstreamer-CUDA-OpenCV pipelines across ARM64/AMD64/X86 architectures and optimized NeRF architectures for parallel multi-GPU processing and camera-Jetson-Unity synchronization

KNOW Corp | *Los Angeles, CA*

January 2023–May 2023

• AI/ML Research Engineer (Co-op)

- Personalized GenerativeQA (GPT-NeoX) using CoT, PEFT LoRA, and Rotary Position Embeddings; compared metric performance (BLEU, ROUGE, STS) and implemented GitHub CI/CD monitoring workflows
- Prepared PoC for 3D Neural Human Avatars with synchronized SpeechT5 Text-to-Speech and deployed via FastAPI for real-time interactive agents

USC Research Labs, USC ISI & IMSC labs | *Marina Del Rey, CA*

January 2022–July 2024

• AI/ML CV Research Engineer & Graduate Assistant

- Published a survey paper NeRFs (Neural Radiance Fields): Past, Present, and Future and applied 3D reconstruction for Astronomy, Geospatial, and Medical domains
- Classified street objects using fine-tuned MobileNet/ViT (83% accuracy (23 ms)), and engineered 3DMM feature regression baselines (62-D) without explicit face landmark detection and I-SPLIT algorithm for edge-devices via modified GradCAM

Sociometrik | *Delhi, India*

October 2020–August 2021

• Data Scientist (GIS, Computer Vision)

- Led Metal Roof-Detection and Super-Resolution pipelines on AWS (EC2/Lambda) using PyTorch Horovod, achieving 0.698 IoU and 0.87 Precision for multi-label terrain classification
- Orchestrated car segmentation and feature extraction workflows using Jenkins CI/CD and RESTful APIs, facilitating automated GIS data analysis from S3 buckets

Indian Institutes of Technology Delhi | *Delhi, India*

January 2020–October 2020

• Research Scientist (Game, Reinforcement Learning)

- Hosted Unity apps for Cybersecurity educational games using seq2seq chatbot & NAF-based actor-critic RL for adaptive NPC
- Implemented user feedback clustering (76% accuracy) and behavioral-tree systems, increasing average user interactivity by 27% based on action-feedback analytics with Firebase storage

TECHNICAL SKILLS

- Languages & frameworks:** Python, C++, Kotlin DSL (CI/CD)
- Libraries:** Pandas, Matplotlib, PySpark, TensorFlow/Keras, PyTorch, ONNX, OpenCV, LangChain, HuggingFace, DBT
- Development expertise:** AWS (EC2, ECS, S3, EMR, Redshift, Lambda, Glue, Athena, Sagemaker), Teamcity, Docker, AirFlow, Kubernetes, Terraform/Terragrunt, PostgreSQL
- ML & GPU Systems Expertise:** NLP (BERT, CLIP, RAG, RLHF), Computer Vision (Diffusion, NeRF, GAN), Computer Graphics, DeepSpeed, TensorRT-LLM, vLLM, Megatron-LM, cuSPARSE, cuBLAS, cuRAND, PTX, OpenGL-CUDA interop

PUBLICATIONS [Scholar]

- Detecting pneumonia using convolutions & dynamic capsule routing...** | *Sensors, MDPI* 2020 (cited: 163)([paper](#))
- Detected: Pneumonia in Chest X-rays; Used: Capsule Networks in conjunction with CNNs; Accuracy Obtained: **96.36% (2.8%↑)**
- Cybersecurity Enhancement through Blockchain Training (CEBT)...** | *IJIM Data Insights, Elsevier* 2021 (cited: 82)([paper](#))
- Developed NAF-based (Actor-Critic Variant) Cont. NPC Adaptiveness Algorithm to augment Game Design & Feedback Mechanics
- Data Augmentation Based Morphological Classification of Galaxies...** | *ESIN, Springer* 2019 (cited: 42)([paper](#))
- Classified: Galaxy Morphologies; Used: Machine Learning, CNNs with L1 and L2 Regularization; Accuracy: 97.92% (0.4%↑)
- NeRFs (Neural Radiance Fields): Past, Present, and Future** | *Arxiv Preprint* 2023 (cited: 30)([paper](#))
- Classified various 3D Computer Vision & Generative AI with History of model-based and image-based Novel View Synthesis
 - Surveying various MPI & NeRF architectures, models, and extensions in terms of objective functions, datasets & evaluation metrics
- AiCNNs (Artificially-integrated Convolutional Neural Networks) for Brain Tumor Prediction** | *EAI* 2019 (cited: 13)([paper](#))
- Classified: 3 types of brain tumors; Used: ML Models, Ensemble CNNs (Regularization); Accuracy: 99.49% (3.72%↑).
- Guess who?-A serious game for cybersecurity professionals** | *9th International Conference, GALA* 2020 (cited: 10)([paper](#))
- Developed a cyber security threat game with clustering based on user feedback (Firebase DB)
- SAVCHOI: Detecting Suspicious Activities using Dense Video Caption...** | *Arxiv Preprint* 2022 (cited: 4)([paper](#))
- Detected: Suspicious activities (Videos); Used: BMT and DETR (Human-Object Interactions) (ResNet50 bb) + Genetic Algo & BERT; BLEU@1: 14.78, BLEU@2: 12.73, BLEU@3: 10.91, BLEU@4: 7.11, METEOR: 16.27, Detection Accuracy: 96.84%
- FuNet-40: fundus disease/abnormality classification using ensemble of fine-tuned pretrained convns** | *T&F'24* (cited: 4)([paper](#))
- Trained 56 models for 15-ensemble models for 40 different fundus diseases/abnormalities; Classification Accuracy: 99.6 %;
 - Visualizations created in 3-D and 2-D for each image using T-SNE and PCA to depict distinct decision boundaries between diseases
- On Multi-Agent Deep Deterministic Policy Gradients and their Explainability for SMARTS Environment** | (cited: 1)([paper](#))
- Compared MAPPO and MADDPG-based RL (with Priority-based Replay Buffers) (on-policy and off-policy RL)

PROJECTS [GitHub]

- CUDAx365: 365 Days of GPU, HPC, and AI Engineering** | *CUDA (PTX/SASS), Triton, LLMs* (CUDAx365) **Jan 2025–Present**
- Mastered GPU internals–developed custom CUDA kernels for memory patterns and warp-level optimizations for rasterization
 - Engineered 2D MHD solvers using MUSCL (achieving near-peak throughput via shared memory padding & graph replays)
- Real-Time CUDA-Q Quantum Error Correction Decoder** | *CUDA-Q, cuBLAS/cuRAND, PyTorch, C++17* **May 2025–Dec 2025**
- Building GPU-accelerated surface-code decoder with CUDA-Q & transformer (2.3× latency reduction vs. CPU baseline)
- Research Paper Implementations (NLP)** | *PyTorch, TensorFlow, DeepSpeed, HuggingFace* ([Gist](#)) **May 2023–March 2024**
- Implemented LoRA (QLoRA, LongQLoRA), Attention, CoT (& ToT), RAG, HNSW for various Transformer architectures
 - Leveraged HF-transformers, bitsandbytes, and Peft for LongNet, LLAMA, RetNet, RWKY
- Multi-Agent Deep Deterministic Policy Gradients and Explainability for SMARTS** (paper) **December 2023–February 2024**
- Proposed a Multi-Agent DDPG with priority-based buffers against top-15 submissions for safer autonomous driving
- Real-Time Data Streaming with Object Detection** | *PySpark, Kafka, StreamLit* (project) **May 2022–June 2022**
- Deployed real-time data pipeline using Kafka to capture and stream data, enabling immediate analysis of incoming information
- Few Shot Learning Approach to Dynamic Intent Satisfaction** | *GPT3, GPT-Neo, NLP* (video) **January 2022–May 2022**
- Proposed LLM-based problem–Intent Satisfaction (fulfill complex intents in slot filling systems) (BLEU (GPT_{dv}): 0.73)
 - Curated function definition and docstrings benchmark dataset of 500,000 tokens for Intent Satisfaction

EDUCATION

- University of Southern California** | *Los Angeles, CA* **August 2021–May 2023**
- **MS in Computer Science**; Courses: Artificial Intelligence, Algorithms, Deep Learning, Applied NLP, Advanced Computer Vision, Autonomous Cyber Physical System, Information Retrieval & Web Search Engines
- Guru Gobind Singh Indraprastha University** | *Delhi, India* **August 2015–May 2019**
- **Bachelors of Technology in Computer Science and Engineering**; Rank: 6; Courses: Operating Systems, Compiler Design, Computer Networks, Artificial Intelligence, Data Mining, Machine Learning (Python), Discrete Mathematics, C++
- Summer School**
- Qiskit Global Summer School (QGSS'21) (certificate) **May 2021–June 2021**
 - Introduction to Quantum Computing (QxQ By Coding School) **October 2020–May 2021**

HONORS & AWARDS

- Viterbi Grad Hack Winners (DeepFake Detection (w/wo LipSync) Chrome Extension) (project) (video) **April 2023**
- Smart India Hackathon'19 3rd Runner-Up (INSURance Recommendations with Integrated Mixed-reality) **April 2019**
- Smart India Hackathon'18 2nd Runner-Up (LOGical Graph Augmented Virtual Indian Map) (project) **April 2018**